Food Inflation Pass-Through to Expectations and Core Inflation: Evidence from India and Brazil

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Abstract

This paper examines whether food-price shocks influence inflation expectations and core inflation in two emerging market economies (India and Brazil). Using a structural vector autoregression (SVAR) with short-run identification we model a transmission channel where exogenous food shocks contemporaneously affect expectations, which then influence core inflation. Empirical impulse-response analysis indicates no statistically significant effect of food-price shocks on expectations or core inflation for India and Brazil. The results are consistent with well-anchored expectations and the strength of inflation-targeting frameworks in these economies.

1 Introduction

Food prices are typically the most volatile component of consumer price indices in many emerging markets, reflecting weather variability, supply-chain frictions and structural features of domestic agricultural markets. In economies where food represents a large share of household consumption, spikes in food prices can produce headline inflation episodes that complicate monetary policy.

Theory suggests that exogenous food-price shocks may alter public inflation expectations; if agents regard shocks as persistent, expectations may shift and feed into core inflation through wage- and price-setting behaviour (so-called second-round effects). Central banks commonly monitor core inflation (excluding food and energy) to separate transitory volatility from underlying trends. The pass-through from food shocks to expectations and core inflation therefore depends both on shock persistence and on the credibility of the monetary policy framework.

This paper focuses on two emerging economies—India and Brazil—for which comparable series on food inflation, survey-based expectations and core inflation are available. We estimate SVAR models with short-run restrictions that reflect the theoretical ordering (food $\rightarrow$ expectations $\rightarrow$ core) and compute impulse-response functions (IRFs) to a one-standard-deviation food-price shock.

The remainder of the paper is organized as follows. Section 2 summarizes the motivating literature. Section 3 sets out the SVAR specification and identification. Section 4 describes the empirical implementation and results. Section 5 concludes. The Appendix provides coefficient tables and IRF figures for India and Brazil.

2 Literature review

The empirical literature on commodity and food-price pass-through documents a large cross-country heterogeneity in the magnitude and persistence of effects on headline and core inflation. While headline inflation in many emerging markets responds strongly to food shocks, evidence on the transmission to survey expectations and to core inflation is mixed, and often depends on the credibility of monetary institutions and the design of inflation measures.

Studies focusing on countries with formal inflation-targeting regimes typically find that well-anchored expectations limit second-round effects. This motivates our focus on whether food shocks contemporaneously move expectations, and if so whether any such movement transmits further into core inflation in the Indian and Brazilian contexts.

3 Model specification

This section describes the reduced-form VAR and the structural representation used to recover economically interpretable shocks. We adopt a short-run identification scheme that reflects the natural ordering: food inflation contemporaneously affects expectations and core inflation, but expectations do not contemporaneously affect food inflation.

3.1 Reduced-form VAR

Define the three-variable vector

$$y_t \;=\; \begin{bmatrix} \pi_t^{\text{food}} \\ E_t\pi_t \\ \pi_t^{\text{core}} \end{bmatrix},$$

where:

- π_t^{food} is the year-over-year food-price inflation rate,
- $E_t\pi_t$ denotes the representative agent's one-period-ahead inflation expectation (survey-based),
- π_t^{core} is core inflation (CPI excluding food and energy).

The reduced-form VAR with p lags is

$$y_t \;=\; A_1 y_{t-1} + A_2 y_{t-2} + \cdots + A_p y_{t-p} + u_t, \qquad u_t \sim \mathcal{N}(0, \Sigma_u), \tag{1}$$

where each A_i is a 3×3 matrix and u_t is the reduced-form residual vector.

3.2 Structural form

The structural form is

$$A_0 y_t \;=\; B_1 y_{t-1} + \cdots + B_p y_{t-p} + \varepsilon_t, \tag{2}$$

where A_0 captures contemporaneous relationships and the structural shocks ε_t are orthogonal with $\mathbb{E}[\varepsilon_t \varepsilon_t'] = I_3$. The mapping between reduced-form residuals and structural shocks is $u_t = A_0^{-1}\varepsilon_t$ and $\Sigma_u = A_0^{-1}(A_0^{-1})'$.

3.3 Short-run identification: matrix form

We impose zero-restrictions on A_0 consistent with the ordering (food $\rightarrow$ expectations $\rightarrow$ core). In matrix form:

$$A_0 = \begin{bmatrix} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{bmatrix}.$$

This structure implies:

1. Food shocks (first equation) are contemporaneously exogenous: expectations and core do not contemporaneously drive food ($a_{12} = a_{13} = 0$).
2. Expectation shocks (second equation) may respond contemporaneously to food but not to core ($a_{23} = 0$).
3. Core-inflation shocks (third equation) may respond contemporaneously to both food and expectations, but do not feed back instantaneously into food or expectations beyond the imposed zeros for identification.

We normalize diagonal elements for scale identification when convenient (e.g., set $a_{11} = a_{22} = a_{33} = 1$) or estimate them as free parameters depending on the chosen estimation routine.

3.4 Estimation and inference

Estimate the reduced-form VAR (1) by OLS to obtain $\{\hat{A}_i\}$ and $\hat{\Sigma}_u$. Recover A_0 by solving $\hat{\Sigma}_u = A_0^{-1}(A_0^{-1})'$ under the zero restrictions (e.g., via maximum likelihood or system-of-equations approaches). Structural shocks follow as $\hat{\varepsilon}_t = A_0\hat{u}_t$. We compute impulse-response functions and bootstrap confidence bands for inference.

4 Empirical results

This section summarizes data, estimation choices and the principal findings for India and Brazil.

4.1 Data and implementation

Monthly (or quarterly where survey frequency dictates) series for food inflation, survey-based one-year-ahead expectations, and core inflation were obtained from established statistical providers. Variables are included in levels as inflation rates; the lag length p is chosen via the Bayesian Information Criterion (BIC). Estimation uses standard OLS for the reduced form and the short-run identification described above.

4.2 Impulse-response analysis

Impulse-response functions (IRFs) trace the dynamic effect of a one-standard-deviation food inflation shock on each variable. For both India and Brazil, the IRFs indicate:

- A pronounced immediate response of headline food inflation to the shock (by construction).
- Little to no statistically significant contemporaneous or persistent response of survey-based inflation expectations.

- Consequently, core inflation shows negligible and statistically insignificant transmission from the food shock in the medium run.

These results are consistent across reasonable alternative lag specifications and bootstrap confidence bands.

4.3 Interpretation

The muted response of expectations in India and Brazil suggests well-anchored inflation expectations, plausibly reflecting central banks' communication and credibility along with the role of core inflation measures in policy frameworks. If agents view food shocks as transitory, they do not update long-run expectations or wage-setting behaviour materially, limiting second-round effects.

4.4 Robustness and limitations

Robustness checks include alternative ordering (e.g., allowing expectations and food contemporaneous interaction), use of different expectation proxies, and subsample analysis. While our main result is stable, data availability for high-frequency expectations constrains some extensions. Future work could integrate market-based expectation measures where available.

5 Conclusion

Using an SVAR with short-run identification for India and Brazil, we find no robust evidence that exogenous food-price shocks materially shift survey-based inflation expectations or transmit into core inflation. This suggests that expectations in these economies are relatively well anchored, and that central banks may not need to respond aggressively to transitory food shocks provided their communications and institutional credibility remain strong.

References

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Appendix

A.1 Estimated coefficient matrices

This subsection presents the reduced-form/identified structural coefficient summaries for India and Brazil. (Estimation details and full output are omitted here; these tables report the matrices used for exposition.)

India

Table A1: SVAR Coefficient Estimates (India)

	inflation_food	expectation	inflation_core
inflation_food	0.051	0	0
expectation	-0.012	1.603	0
inflation_core	0	0	0.321

Brazil

Table A2: SVAR Coefficient Estimates (Brazil)

	inflation_food	expectation	inflation_core
inflation_food	0.069	0	0
expectation	0.004	3.331	0
inflation_core	0	0	0.369

A.2 Impulse-response functions (IRFs)

Below are the IRF figures for India and Brazil. Ensure the image files `India.png` and `Brazil.png` are placed in the same folder as this `.tex` file (Overleaf is case-sensitive).

India

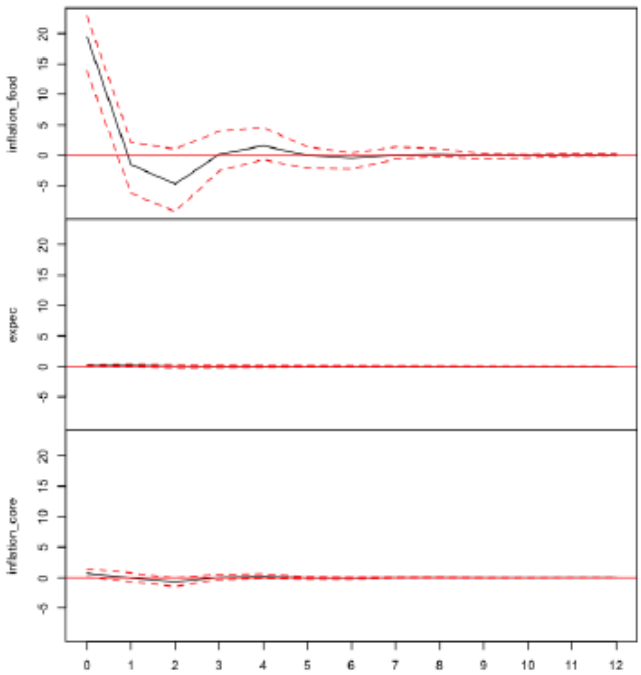

Figure 1: Impulse Response Function: India

Figure 1: Impulse Response Function: India

Brazil

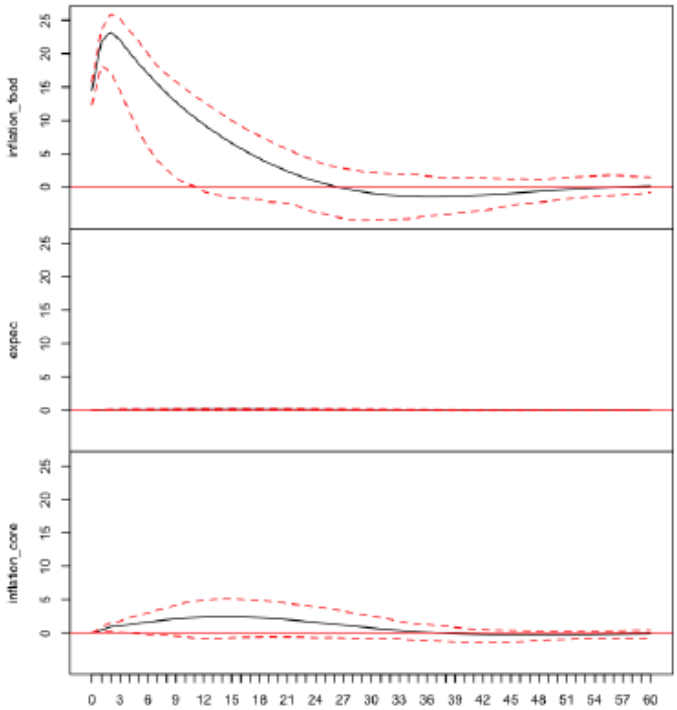

Figure 2: Impulse Response Function: Brazil

Figure 2: Impulse Response Function: Brazil

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